

SC3102: Signals, systems and transforms

Tutorial 1: Signals 1 (Classification of signals, autocorrelation)

1. Prove or disprove whether the following signals are periodic using the definitions of periodicity. If the signals are periodic, determine its fundamental period.
 - (a) $x(t) = e^{3.1t}$
 - (b) $x(t) = e^{j3.1t}$, where $j = \sqrt{-1}$
 - (c) $x(t) = \sum_{n=-\infty}^{+\infty} g(t - 3n)$ where $g(t)$ is an arbitrary function of time
2. Check if the signals below are periodic. For periodic signals, find the fundamental period. Assume the time is measured in seconds.
 - (a) $x(t) = \cos(20t + 1) + \sin(30\pi t - 4)$
 - (b) $x(t) = \cos^2(10t)$
3. Classify the following signals as energy signal or as power signal.
 - (a) $x(t) = e^{j\pi t} \quad -\infty < t < \infty$
 - (b) $x(t) = tu(t)$ where $u(t)$ is a unit step function
4. Consider two orthogonal energy signals $x(t)$ and $y(t)$ with energies E_x and E_y , respectively. Obtain the energy of the signal $x(t) + y(t)$ and show that it is identical to the energy of the signal $x(t) - y(t)$. Comment about the power of the resulting signals.
Note: If $x(t)$ and $y(t)$ are orthogonal, $\int_{-\infty}^{\infty} x(t)y(t)dt = 0$
5. Which of the following functions are satisfying the properties of an autocorrelation function? Justify your answers.
 - (a) $x(\tau) = \delta(\tau) + \sin(2\pi f_1 \tau)$
 - (b) $x(\tau) = \begin{cases} 1 - |\tau| & -1 \leq \tau \leq 1 \\ 0 & \text{elsewhere} \end{cases}$

Note the following Fourier transform relations:

$$F\left[\text{tri}\left(\frac{t}{T}\right)\right] = T \text{sinc}^2(fT) \quad \text{where } \text{tri}(\) \text{ is a triangular function and } \text{sinc}(t) = \frac{\sin(\pi t)}{\pi t} \text{ is a sinc function.}$$

Answers:

1. (a) Non-periodic.
(b) Periodic, Fundamental period $T_0 = \frac{2\pi}{3.1}$.
(c) Periodic, Fundamental period $T_0 = 3$.
2. (a) Non-periodic.
(b) Periodic, Fundamental period $T_0 = \frac{\pi}{10}$
3. (a) Power signal.
(b) Neither energy signal, nor power signal.
4. Energy of $x(t) + y(t)$: $E_{x+y} = E_x + E_y$
5. (a) Not satisfying the properties of autocorrelation function
(b) It is an autocorrelation function

Exercises:

1. Prove or disprove whether the following signals are periodic using the definitions of periodicity. If the signals are periodic, determine its fundamental period.
(a) $x(t) = \cos(20t + 1)\sin(30t - 4)$
(b) $x(t) = g(2t)$, where $g(t)$ is periodic with period T_0

2. Classify the following signal as energy signals or as power signal.

$$x(t) = \begin{cases} A \cos(2\pi f_0 t) & -\frac{T_0}{2} \leq t \leq \frac{T_0}{2}, \text{ where } T_0 = \frac{1}{f_0} \\ 0 & \text{elsewhere} \end{cases}$$

3. Check whether the function $x(\tau) = \begin{cases} 1 & -1 \leq \tau \leq 1 \\ 0 & \text{elsewhere} \end{cases}$ satisfies the properties of an autocorrelation function? Justify your answers.

Note: $F\left[\Pi\left(\frac{t}{T}\right)\right] = T \text{sinc}(fT)$ where $\Pi(\quad)$ is a rectangular function.

Answers:

1. (a) Periodic, Fundamental period $T_0 = \frac{\pi}{5}$
(b) Periodic, Fundamental period $\frac{T_0}{2}$.
 2. Energy signal.
 3. It is not an autocorrelation function since power spectral density is a positive and negative going function.
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